

GARP Risk & AI — Memory Flashcards

Names, Theories, Formulas, Acronyms — 150 Slides

Unofficial

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Formula

$$d_E = \sqrt{\sum_{j=1}^m (x_{jQ} - x_{jP})^2}$$

Memory Hook

L2 norm. “As the crow flies.” Used in K-means by default. Squares differences so large deviations are penalized heavily. Best when clusters are spherical and features are on similar scales. Sensitive to outliers because of the squaring.

Formula

$$d_M = \sum_{j=1}^m |x_{jQ} - x_{jP}|$$

Memory Hook

L1 norm. "City block driving." More robust to outliers than Euclidean because it does not square differences. Used when features have different scales or when the data has heavy tails. Also called taxicab distance.

Formula

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

Memory Hook

a_i = mean distance to own cluster (cohesion). b_i = mean distance to nearest other cluster (separation). Range $[-1, 1]$. $s \approx 1$ = well-clustered. $s \approx 0$ = boundary. $s < 0$ = mis-clustered. Used to select optimal K in K-means.

Formula

$$WCSS_k = \sum_{j=1}^{n_k} d_j^2$$

Memory Hook

Also called inertia. K-means objective function. Sum of squared distances from each point to its cluster centroid. Lower = more compact clusters. WCSS always decreases as K increases (overfitting risk).

Formula

$$WCSS = \sum_{k=1}^K WCSS_k$$

Memory Hook

Sum of all cluster WCSS values. K-means minimizes this. Used in scree plot: plot WCSS vs K, look for elbow. Elbow = point where adding more clusters yields diminishing returns.

Formula

$$BCSS = \sum_{k=1}^K n_k \|\mu_k - \bar{\mu}\|^2$$

Memory Hook

Measures how far cluster centroids are from the overall mean, weighted by cluster size. Higher = better separated clusters. Relationship: Total Variance = WCSS + BCSS.

Formula

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Memory Hook

Underfitting = high bias, low variance. Overfitting = low bias, high variance. Irreducible error = noise no model can eliminate. Trade-off: increasing complexity reduces bias but increases variance.

Formula

$$L = \frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$$

Memory Hook

L2 penalty. Shrinks coefficients toward zero but never exactly zero. Good when many features are correlated. λ controls strength: larger λ = more shrinkage. Does not perform feature selection.

Formula

$$L = \frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$$

Memory Hook

L1 penalty. Can set coefficients exactly to zero = feature selection. Good when only a few features are important. λ controls sparsity: larger λ = more coefficients zeroed out.

Formula

$$L = \frac{1}{N} \sum (y_i - \hat{y}_i)^2 + \lambda_1 \sum \beta_j^2 + \lambda_2 \sum |\beta_j|$$

Memory Hook

Combines L1 and L2 penalties. Gets sparsity from LASSO and grouping from Ridge. Good when features are correlated and you want some feature selection. Two hyperparameters to tune.

Formula

$$\text{Precision} = \frac{TP}{TP+FP}$$

Memory Hook

Of those predicted positive, how many are actually positive? High precision = few false alarms. Use when false positives are costly (spam filter, fraud alert). Also called positive predictive value.

Formula

$$\text{Recall} = \frac{TP}{TP+FN}$$

Memory Hook

Of all actual positives, how many did we catch? High recall = few missed positives. Use when false negatives are costly (fraud detection, medical diagnosis). Also called true positive rate.

Formula

$$\text{Specificity} = \frac{TN}{TN+FP}$$

Memory Hook

True negative rate. Of all actual negatives, how many did we correctly identify? Important when false positives are costly. Complement of false positive rate: $\text{Specificity} = 1 - \text{FPR}$.

Formula

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Memory Hook

Harmonic mean of precision and recall. Range [0,1]. Balances both metrics. Low F1 can be from poor precision, poor recall, or both. Useful for imbalanced datasets.

Formula

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Memory Hook

Percentage of correct predictions. Misleading for imbalanced data. If 98% of transactions are legitimate, predicting all legitimate gives 98% accuracy but 0% fraud detection. GARP tests this concept frequently.

Formula

$$\text{Error Rate} = \frac{FP+FN}{\text{Total}} = 1 - \text{Accuracy}$$

Memory Hook

Percentage of incorrect predictions. Same limitations as accuracy for imbalanced data. Type I error = false positive. Type II error = false negative. Costs of each type differ by application.

Formula

$$MSE = \frac{1}{N} \sum (y_i - \hat{y}_i)^2$$

Memory Hook

Mean Squared Error. Penalizes large errors heavily because of squaring. Sensitive to outliers. Units are squared (e.g., dollars squared), making interpretation difficult.

Formula

$$RMSE = \sqrt{MSE}$$

Memory Hook

Root Mean Squared Error. Same units as target variable, so more interpretable. Still sensitive to outliers. ECB recommends reporting RMSE alongside MAE for tail risk.

Formula

$$MAE = \frac{1}{N} \sum |y_i - \hat{y}_i|$$

Memory Hook

Mean Absolute Error. Robust to outliers. Straightforward interpretation: average error magnitude. Does not penalize large errors as heavily as MSE. Can mask direction of bias.

Formula

$$MAPE = 100 \times \frac{1}{N} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Memory Hook

Mean Absolute Percentage Error. Scale-independent, good for comparing across different scales. Fails when actual = 0 or near zero. Board-friendly because it is a percentage.

Formula

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Memory Hook

β_0 = intercept (value of y when $x = 0$). β_1 = slope (change in y for one-unit change in x). u_i = error term. OLS minimizes sum of squared residuals.

Formula

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_m x_{mi} + u_i$$

Memory Hook

m features, $m + 1$ parameters. Each coefficient measures partial effect controlling for other features. OLS can be used if model is linear in parameters. Watch for multicollinearity, heteroskedasticity, outliers.

Formula

$$f(z) = \frac{1}{1+e^{-z}}$$

Memory Hook

Sigmoid function. Output between 0 and 1. Used in logistic regression for binary classification. Also used as activation function in neural networks. Can suffer from vanishing gradient.

Formula

$$P_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{1i} + \dots + \beta_m x_{mi})}}$$

Memory Hook

Applies sigmoid to linear combination of features. Output interpreted as probability. Log-odds is linear: $\ln(P/(1 - P)) = \beta_0 + \beta_1 x_1 + \dots$. Coefficients interpreted as odds ratios: $\exp(\beta_j)$.

Formula

$$\log L = \sum [y_i \ln F(y_i) + (\mathbf{1} - y_i) \ln(\mathbf{1} - F(y_i))]$$

Memory Hook

Sum of log-probabilities. MLE maximizes this. Log transformation turns products into sums, avoiding underflow. For linear regression with normal errors, MLE = OLS.

Formula

$$\hat{\beta}_1 = \frac{\sum y_i x_i - N \bar{y} \bar{x}}{\sum x_i^2 - N \bar{x}^2}$$

Memory Hook

Covariance(x,y) / Variance(x). Measures how much y changes per unit change in x. OLS finds the line that minimizes sum of squared residuals.

Formula

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Memory Hook

Regression line passes through the point of means $(\bar{x}, \bar{y})$. Intercept = predicted y when $x = 0$. May not be meaningful if $x = 0$ is outside data range.

Formula

$$RSS = \sum (y_i - \hat{y}_i)^2$$

Memory Hook

Residual Sum of Squares. OLS minimizes this. Also called Sum of Squared Residuals (SSR) or Sum of Squared Errors (SSE). Squaring prevents positive and negative residuals from canceling.

Formula

$$z = \frac{x - \mu}{\sigma}$$

Memory Hook

Z-score. Mean = 0, SD = 1. Preferred when data has outliers (normalization would squash them). Applied to input features, not target. Critical for K-means, KNN, SVM, neural networks.

Formula

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Memory Hook

Range [0,1]. Squashes outliers into tight range. Preserves correlations between data points. Does not change shape of distribution. Less preferred than standardization when outliers exist.

Formula

$$PC_j = b_{1j}P_1 + b_{2j}P_2 + \cdots + b_{mj}P_m$$

Memory Hook

Linear combination of original predictors. Weights = component loadings. First PC captures most variance. Second PC orthogonal to first. Components are uncorrelated. Used for dimensionality reduction and multicollinearity.

Formula

$$S_c(P, Q) = \frac{P \cdot Q}{\|P\| \|Q\|}$$

Memory Hook

Range $[-1, 1]$. Measures angle between vectors, not magnitude. Used in Word2Vec for word similarity. If words are close, similarity near 1. If unrelated, similarity near 0. Example: China-Japan = 0.84.

Formula

$$TF\text{-}IDF_{ij} = TF_{ij} \times IDF_i$$

Memory Hook

Term frequency $\times$ inverse document frequency. Weights words that are frequent in a document but rare across corpus. Words appearing in all documents get $IDF = 0$. Used for document classification and information retrieval.

Formula

$$IDF_i = \ln \left(\frac{D}{d_i} \right)$$

Memory Hook

D = total documents. d_i = documents containing term i . Rare words get higher IDF. Common words get lower IDF. If term appears in all documents, $IDF = 0$ (no discriminatory power).

Formula

$$v_{norm} = \frac{v}{\sqrt{\sum v_i^2}}$$

Memory Hook

Creates unit vector. Divides each element by L2 norm. Used in NLP to normalize word vectors. Prevents repeated words from skewing analysis. All elements lie between 0 and 1.

Formula

$$P(c|d) = \frac{P(d|c)P(c)}{P(d)}$$

Memory Hook

Posterior = Likelihood × Prior / Evidence. $P(c)$ = prior probability. $P(d|c)$ = likelihood. $P(c|d)$ = posterior. Used in Naive Bayes classification.

Formula

$$\hat{c} = \arg \max_c P(w_1|c)P(w_2|c) \cdots P(w_N|c)P(c)$$

Memory Hook

Product of conditional word probabilities times class prior. “Naive” because assumes word independence. Laplace smoothing adds 1 to all counts to avoid zero probabilities. Used for spam detection, sentiment analysis.

Formula

$$\text{score}(Q_i, K_j) = \frac{Q_i \cdot K_j}{\sqrt{d_k}}$$

Memory Hook

Divide by $\sqrt{d_k}$ (key dimension) to prevent softmax saturation. Query = current token. Key = other tokens. Value = used to compute context. Core of transformer architecture.

Formula

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

Memory Hook

Normalizes exponentials into probability distribution. Sum = 1. Used in attention and multiclass outputs. Exponentially magnifies largest values. Prevents negative probabilities.

Formula

$$w_i^{new} = w_i^{old} - \eta \frac{\partial L}{\partial w_i}$$

Memory Hook

Subtract learning-rate-scaled gradient. η = learning rate. Too small = slow convergence. Too large = overshooting. Batch GD uses full dataset. SGD uses one example. Mini-batch uses subset.

Formula

$$\eta_t = \eta_0 e^{-kt}$$

Memory Hook

Learning rate decreases exponentially over epochs. k controls decay rate. t = epoch. Start with larger rate for fast initial progress, reduce for fine-tuning. Common in SGD.

Formula

$$\eta_t = \frac{\eta_0}{1+kt}$$

Memory Hook

Alternative decay schedule. Decreases more slowly than exponential. k controls decay rate. Used in stochastic gradient descent. Both schedules reduce learning rate over time.

Formula

$$Gini = 1 - \sum_{j=1}^J p_j^2$$

Memory Hook

Node impurity in classification trees. Lower = purer. Gini = 0 for pure node. Gini = 0.5 for binary 50/50 split. Computationally faster than entropy (no log). Usually produces similar trees.

Formula

$$\text{Entropy} = - \sum_{j=1}^J p_j \log_2(p_j)$$

Memory Hook

Measure of disorder in classification trees. Lower = purer. Entropy = 0 for pure node. Entropy = 1 for binary 50/50 split. Base-2 log. Changing base scales all values by constant.

Formula

$$AIC = 2k - 2 \ln(L)$$

Memory Hook

Akaike Information Criterion. k = parameters. L = likelihood. Lower = better. Penalizes complexity. Used in stepwise regression for feature selection. Does not require nested models.

Formula

$$BIC = k \ln(N) - 2 \ln(L)$$

Memory Hook

Bayesian Information Criterion. N = sample size. Penalizes complexity more than AIC for large samples. Lower = better. More conservative (selects simpler models). Used for model selection.

Formula

$$\text{Cov}(x_1, x_2) = \frac{\sum (x_1 - \bar{x}_1)(x_2 - \bar{x}_2)}{N-1}$$

Memory Hook

Average product of deviations from means. Positive = variables move together. Negative = move oppositely. Zero = no linear relationship. Used in LDA covariance matrix.

Formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Memory Hook

Swap diagonal, negate off-diagonal, divide by determinant. Determinant = $ad - bc$ must be non-zero. Used in LDA discriminant function. If determinant = 0, matrix is singular (no inverse).

Formula

$$\delta_j(x) = x^T \hat{\Sigma}^{-1} \hat{\mu}_j - \frac{1}{2} \hat{\mu}_j^T \hat{\Sigma}^{-1} \hat{\mu}_j + \ln(\hat{\pi}_j)$$

Memory Hook

LDA assumes multivariate normal with common covariance. Assign to class with largest δ_j . Works for multi-class classification. $\hat{\pi}_j$ = prior probability of class j .

Formula

$$L = \sum_{j=1}^m \sum_{i=1}^N (x_{ij} - \hat{x}_{ij})^2$$

Memory Hook

Reconstruction error. Autoencoders learn compressed representations. Used for dimensionality reduction and anomaly detection. High reconstruction error = anomaly. With linear activation = PCA.

Contribution

Coined “Artificial Intelligence” (1956)

Memory Hook

Dartmouth Conference. Along with Marvin Minsky, Nathan Rochester, and Claude Shannon. Also developed LISP programming language (1960). Founded Stanford AI Lab (1963). Considered a founding father of AI.

Contributions

Turing Machine (1936). Turing Test (1950) = "The Imitation Game."

Memory Hook

"On Computable Numbers, with an Application to the Entscheidungsproblem" (1936). "Computing Machinery and Intelligence" (1950). Turing Test: computer proves intelligence if conversation indistinguishable from human. Worked at Bletchley Park during WWII.

Contribution

First neural network model (1943)

Memory Hook

"A logical calculus of the ideas immanent in nervous activity." Published in Bulletin of Mathematical Biology. Described neurons in terms of propositional logic. Foundational paper for neural networks.

Contribution

Hebbian Learning (1949)

Memory Hook

“The Organisation of Behaviour.” “Neurons that fire together, wire together.” Proposed associative learning in neural networks. Neurophysiological postulate: connection strengthens when one neuron repeatedly fires another.

Contribution

Perceptron (1958)

Memory Hook

Formulated Perceptron in terms of probability theory, not symbolic logic. Single-layer neural network. Could not learn XOR (shown by Minsky and Papert). Inspired widespread interest in neural networks.

Contribution

"Perceptrons" (1969)

Memory Hook

Showed single-layer Perceptrons cannot learn simple logical functions like XOR. Critique severely undermined enthusiasm for neural networks. Later shown that multi-layer networks can learn XOR. Led to AI winter.

Contribution

"Parallel Distributed Processing" (1986)

Memory Hook

Revived connectionism. Demonstrated how layers of interacting neurons could learn complex functions. Displaced pessimism from Minsky and Papert. Backpropagation rose to prominence.

Contribution**LeNet-5 (1998)****Memory Hook**

Convolutional neural network for digit recognition. 8 layers with convolutional and pooling layers. About 9,118 neurons. Foundational work for modern CNNs. Used for mail sorting, bank check processing.

Contribution**AlexNet (2012)****Memory Hook**

Won ImageNet Large Scale Visual Recognition Challenge with 15.3% error rate (vs 25.8% previous year). Nearly 100 times bigger than LeNet-5. Almost 900,000 neurons. Demonstrated power of deep learning. Used GPUs for computation.

Contributions

AlphaGo (2016): defeated Lee Sedol 4-1. AlphaZero (2017): self-play.

Memory Hook

AlphaGo beat Fan Hui 5-0 (2015), Lee Sedol 4-1 (2016). AlphaZero learned by competing against itself without human game databases. Defeated Stockfish. Acquired by Google in 2014.

Contribution

"Stochastic parrot" (2021)

Memory Hook

Describes LLMs that imitate language without understanding meaning. They generate plausible responses based on statistical patterns, not true understanding. Term widely cited in AI ethics discussions.

Contribution

"Attention Is All You Need" (2017)

Memory Hook

Introduced transformer architecture based on self-attention. Eliminated need for recurrence and convolutions. Foundation for BERT, GPT, and modern LLMs. Multiple attention heads capture different relationships.

Contribution**BERT (2018)****Memory Hook**

Bidirectional Encoder Representations from Transformers. Encoder-only model. 100 million parameters. Trained on books and Wikipedia. Not generative. Excellent for classification. Variants: RoBERTa, FinBERT.

Contribution

GPT series (2018-present)

Memory Hook

Generative Pre-trained Transformer. Decoder-only autoregressive model. GPT-1: 100M parameters. GPT-2: 774M. GPT-3: 175B. GPT-4: 2023. GPT-5: 2025. Uses RLHF for alignment.

Contribution**BART (2019)****Memory Hook**

Bidirectional and Auto-Regressive Transformer. Combines encoder and decoder. Used for summarization, translation, and generation. Trained by distorting text and learning to reconstruct. Open source.

Contribution

Fairness Impossibility Theorem (2016)

Memory Hook

Demographic parity, predictive rate parity, and equal opportunity cannot all hold simultaneously when base rates differ. Mathematical result, not practical limitation. Fairness requires trade-offs and value judgments.

Contribution**Utilitarianism****Memory Hook**

Jeremy Bentham and John Stuart Mill. Maximize overall utility/happiness. Consequentialist framework. Judges actions by outcomes, not intentions. Can justify violating individual rights for greater good.

Contribution

Deontology. Categorical Imperative.

Memory Hook

Act only on maxims that could be universal laws. Treat people as ends, not means. Judges actions by adherence to duties and rules, not consequences. Lying is always wrong regardless of consequences.

Contribution

Virtue Ethics

Memory Hook

Nicomachean Ethics. Emphasizes virtuous character traits (wisdom, courage, justice, temperance). Asks “What kind of person should I be?” rather than “What should I do?” Virtues nurtured through practice and habit.

Contribution

Lloyd's Algorithm (1957) = K-means

Memory Hook

Bell Labs. Iterative algorithm: assign points to nearest centroid, update centroids, repeat until convergence. Also called K-means. K-means++ improves initialization by spreading centroids apart.

Statement

Neural network with enough hidden neurons can approximate any continuous function.

Memory Hook

Power of neural networks. Does not guarantee learning or generalization. Need enough data and proper training. Underlies why neural networks are so flexible. Requires non-linear activation functions.

Statement

When base rates differ, demographic parity, predictive rate parity, and equal opportunity cannot all hold.

Memory Hook

Base rate = prevalence of outcome in group. Fairness requires trade-offs. Must choose which metric matters most. Kleinberg, Mullainathan, Raghavan (2016). Chouldechova independently derived related result.

Statement

No single algorithm is best for all problems.

Memory Hook

Match algorithm to data characteristics. K-means for spherical clusters. DBSCAN for arbitrary shapes. Neural networks for complex patterns. This is why GARP covers so many algorithms.

Statement

Total error = Bias² + Variance + Irreducible Error.

Memory Hook

Simple = high bias (underfitting). Complex = high variance (overfitting). Irreducible error = noise. Regularization, cross-validation, ensembles help find balance. Central tension in machine learning.

Statement

As features increase, data becomes sparse. Distances become similar.

Memory Hook

K-means struggles with many features. KNN also affected. Solutions: feature selection, PCA, cosine distance. Distances between points become less meaningful. Need exponentially more data as dimensions increase.

Statement

Gradients become very small in deep networks. Learning slow or impossible.

Memory Hook

Caused by sigmoid/tanh activation. Chain rule multiplies many small derivatives. Solutions: ReLU activation, batch normalization, LSTM for RNNs, residual connections. More critical for deep networks.

Statement

Gradients grow extremely large. Unstable training.

Memory Hook

Opposite of vanishing gradient. Chain rule multiplies many large derivatives. Solutions: gradient clipping, careful weight initialization, batch normalization. Less common than vanishing gradient.

Definition

Model captures noise. Good on training, poor on test.

Memory Hook

Large gap between training and test error. Signs: very low training error, high test error. Solutions: regularization (L1, L2), dropout, early stopping, more data, simpler model. More severe with many parameters.

Definition

Model too simple. Poor on both training and test.

Memory Hook

High bias. More complex model needed. Signs: high training error, high test error. Solutions: add features, use more complex model, reduce regularization. Less common than overfitting.

Definition

States, actions, rewards, transition probabilities. Markov property: no memory.

Memory Hook

Bellman equations define optimal value functions. Dynamic programming solves MDPs when probabilities known. Foundation of reinforcement learning. Future depends only on current state.

Formula

$$V^*(s) = \max_a E[R_{t+1} + \gamma V^*(s')]$$

Memory Hook

Optimal value = max expected reward + discounted next value. γ = discount factor (0 to 1). Higher γ = more weight on future rewards. Used in Q-learning and dynamic programming.

Definition

Explore: try new actions. Exploit: use known best.

Memory Hook

Epsilon-greedy balances both. With probability ϵ , explore randomly. With probability $1 - \epsilon$, exploit best-known action. ϵ typically decays over time. Pure greedy may miss better options.

Comparison

MC: updates at episode end. TD: updates each step.

Memory Hook

MC high variance, unbiased. TD lower variance, biased. MC requires complete episodes. TD works for continuous tasks. Q-learning is TD method. TD more commonly used in practice.

Formula

$$Q(s, a) \leftarrow Q(s, a) + \alpha[R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Memory Hook

Off-policy TD. α = learning rate. γ = discount factor. Learns optimal action-value function. DQN combines with neural networks. Uses experience replay and target network.

Definition

Finds linear combinations maximizing variance.

Memory Hook

First PC most variance. Second PC orthogonal to first. Components uncorrelated. Used for dimensionality reduction and multicollinearity. Not a learning method (no training). Sensitive to scaling.

Comparison

PCA: linear. Autoencoder: nonlinear with activation functions.

Memory Hook

Autoencoder with linear activation = PCA. Nonlinear captures more complex patterns. Used for anomaly detection. High reconstruction error = anomaly. Bottleneck layer = compressed representation.

Comparison

Bagging: bootstrap samples. Random Forest: bagging + random feature subsetting.

Memory Hook

RF reduces correlation between trees. Key hyperparameter: number of features at each split (typically $\sqrt{m}$). More diverse trees = better ensemble. Reduces variance. Less interpretable than single tree.

Comparison

Bagging: parallel. Boosting: sequential, corrects errors.

Memory Hook

Bagging reduces variance. Boosting reduces bias. AdaBoost reweights misclassified examples. Gradient Boosting fits residuals. Boosting can overfit if not tuned carefully. XGBoost is popular implementation.

Definition

Maximum margin hyperplane. Support vectors = closest points.

Memory Hook

Kernel trick for nonlinear. Soft margin allows misclassification. λ controls trade-off: large λ = wider margin, more misclassifications. Works well with high dimensions. Binary classification primarily.

Definition

Classifies based on majority vote of K nearest neighbors.

Memory Hook

Lazy learner (no training). $K = \sqrt{N}$ common heuristic. Small K = overfitting. Large K = underfitting. Requires feature scaling. Computationally intensive at prediction time.

Definition

Recursive binary splitting. Interpretable ("white box").

Memory Hook

Regression: minimize RSS. Classification: minimize Gini/entropy. Pruning prevents overfitting. Pre-pruning stops growth early. Post-pruning grows fully then removes weak branches. Ensemble methods improve accuracy.

Definition

Inability to understand internal mechanisms.

Memory Hook

Distinct from explainability (specific decisions) and interpretability (internal logic). Deep networks are inscrutable. Leads to hidden bias, privacy risks, over-reliance. Source of reputational risk.

Definition

Ability to explain specific decisions in human terms.

Memory Hook

Post-hoc (after the fact). Counterfactual explanations: “If you had X, you would have been approved.” Techniques: SHAP, LIME, feature importance. Builds trust and accountability.

Definition

Degree to which human can comprehend internal logic.

Memory Hook

Inherent (built-in). Decision trees = interpretable. Neural networks = not interpretable. Trade-off with accuracy. Less flexible models more interpretable.

Definition

Openness about design, algorithms, data, processes.

Memory Hook

Distinct from explainability and interpretability. Transparency = knowing what the model is. Interpretability = understanding how it works. Explainability = explaining specific decisions.

Four Sources

Problem specification, data, modeling, deployment.

Memory Hook

Proxy discrimination: protected attribute removed but correlated features remain. Historical bias: training data reflects past discrimination. Sampling bias: sample not representative. Measurement bias: inconsistent measurement across groups.

Definitions

Demographic parity: equal positive prediction rates. Predictive rate parity: equal precision. Equal opportunity: equal TPR.

Memory Hook

Impossibility theorem: cannot satisfy all three. Individual fairness: treat similar individuals similarly. Group fairness: statistical parity across groups. Trade-offs required.

Three Frameworks

Consequentialism (outcomes). Deontology (duties). Virtue ethics (character).

Memory Hook

Complementary, not mutually exclusive. Practical ethics blends all three. Best decision accords with all three. Medical ethics provides model for AI ethics.

Five Principles

Nonmaleficence, beneficence, justice, autonomy, explainability.

Memory Hook

Nonmaleficence = do no harm. Beneficence = do good. Justice = fairness. Autonomy = respect decisions. Explainability = understandable reasons. From medical ethics (Hippocratic Oath, Belmont Report).

Key Principles

Data minimization, right to be forgotten, contextual integrity, control over data, data security.

Memory Hook

GDPR Article 17 = right to erasure. Article 33 = breach notification within 72 hours. Privacy is both moral and legal right. Data minimization = collect only what is necessary. Contextual integrity = respect context norms.

Key Concepts

Model inventory, model landscape, model risk appetite, effective challenge.

Memory Hook

SR 11-7 requires annual validation at minimum. AI/ML models require more frequent monitoring. Three lines of defense: business, risk management, internal audit. Model inventory = registry. Model landscape = dependency map.

Compared to Traditional Models

More frequent monitoring. Broader stakeholder involvement. Focus on explainability and benchmarking.

Memory Hook

Black-box models require different validation. K-fold cross-validation more suitable than MSE. Explainability, benchmarking, assessment on different datasets, back testing. HR, compliance, operational risk involved.

Key Risks

Hallucination, unpredictability, data privacy, IP issues, bias amplification.

Memory Hook

Hallucination = factually incorrect but convincing. Unpredictability = same prompt, different outputs. Data privacy = prompts may contain sensitive data. IP = training on copyrighted content. Bias = amplifies training data biases.

Key Regulations

EU AI Act, GDPR, NIST AI RMF, SR 11-7.

Memory Hook

EU AI Act fines up to 35M euros or 7% of global turnover. Prohibited: social scoring, biometric categorization, emotion recognition in workplace. High-risk: impact assessment required. NIST AI RMF: Govern, Map, Measure, Manage.

Prohibited

Social scoring, biometric categorization, emotion recognition in workplace, untargeted facial scraping, manipulation of free will.

Memory Hook

Effective August 1, 2024. Ban on unacceptable risk from February 2, 2025. Most provisions from August 2, 2026. European AI Office enforces.

Requirements

Impact assessment, record keeping, transparency, human oversight.

Memory Hook

Banks, insurance, essential services. Must keep records of datasets, methodologies, oversight measures. Impact assessment on fundamental rights. Notify when interacting with chatbots/biometric/emotion recognition.

Articles

Article 17 = right to erasure. Article 33 = breach notification (72 hours). Article 6 = lawful basis. Article 22 = automated decision-making.

Memory Hook

Extraterritorial scope: applies to organizations outside EEA offering services to EEA data subjects. Data subject rights: access, rectification, erasure, restriction, portability, objection.

Four Functions

Govern, Map, Measure, Manage.

Memory Hook

Voluntary, non-prescriptive, flexible. Govern = structures. Map = context. Measure = assessment. Manage = mitigation. Customizable based on use cases and risk profiles.

Key Points

Federal Reserve and OCC. Model risk management guidance. Annual validation minimum.

Memory Hook

Defines model as quantitative method/system/approach. Effective challenge = critical analysis by independent parties. Three lines of defense. Model inventory and landscape required.

Lines

First: business/model owners. **Second:** independent risk management. **Third:** internal audit.

Memory Hook

First line owns model and performance. **Second line** sets policy, conducts validations. **Third line** provides independent assurance to board. IIA formalized model.

Contents

Owner, purpose, methodology, risk tier, validation status, interdependencies.

Memory Hook

Registry of all models. Includes AI/ML models. Tracks usage, changes, approvals. Model landscape = map of interactions. Identifies cascade risk and critical nodes.

Definition

Acceptable level of model risk. Set by board.

Memory Hook

Guides governance. Model risk function sets thresholds for materiality. Board sets overall appetite. Model risk ranking: materiality, complexity, exposure.

Definition

Qualitative assessment of whether model is used appropriately.

Memory Hook

People-focused, not numbers-focused. Assesses acceptance, trust, comprehensibility. Results presented to senior management. Contributes to risk modeling culture.

Definition

Compare model predictions against realized outcomes.

Memory Hook

Detects bias, degradation, incorrect assumptions. Part of ongoing validation. Compares predicted vs actual. Backtesting at appropriate cadence.

Three Goals

Operational feasibility, proper documentation, user training.

Memory Hook

Ensures model can run as intended. Adheres to firm-wide standards. Users trained to interpret results. Preproduction phase required. Results form basis for approval.

Definition

Critical analysis by objective and informed parties.

Memory Hook

SR 11-7 concept. Identifies assumptions, limitations, weaknesses. Requires independence, competence, influence. Core of model validation.

Frequency

At least annually, or more frequently if warranted.

Memory Hook

SR 11-7 recommendation. AI/ML models need more frequent review. Higher-risk models validated more often. Periodic baseline revalidation. Change-based validation for material changes.

Definition

Determines depth and frequency of validation.

Memory Hook

Higher tier = more frequent validation. Based on materiality, complexity, exposure. Drives resource allocation. Minimum standards apply to all models.

Factors

Materiality, complexity, exposure.

Memory Hook

Materiality = impact on decisions. Complexity = methodology sophistication. Exposure = usage breadth. Used to prioritize validation resources.

Definition

Origin, history, and legal right to use data.

Memory Hook

Critical for alternative data. FTC requires destruction of models built on improperly obtained data. Copyright lawsuits against GenAI vendors. Vendor models need proof of ownership.

Definition

Categorizing data by sensitivity and structure.

Memory Hook

Sensitivity labels: public, internal, confidential, restricted. Structure labels: structured, semi-structured, unstructured. Determines handling and controls.

Definition

Data about data.

Memory Hook

Documents origin, structure, definitions, relationships. Types: descriptive, structural, administrative, reference, statistical. Essential for robust data governance.

Definition

Role-Based Access Control.

Memory Hook

Permissions based on organizational roles. ABAC = attribute-based. MAC = mandatory. DAC = discretionary. RBAC simpler; ABAC more flexible.

Definition

Attribute-Based Access Control.

Memory Hook

Uses attributes of user, resource, environment. More flexible than RBAC. Can implement complex policies. More complex to manage.

Elements

Vision, goals, priorities, data governance policy.

Memory Hook

Includes authorization for non-public data. Single or recurring use. Source of information. Alternative data requires additional controls.

Responsibilities

Approved data practices and policies.

Memory Hook

Collection, reconciliations, treatment, access, monitoring. Management of data stewards. Overall oversight, not operational. Data strategy approval.

Purpose

Augment available data and mitigate privacy risks.

Memory Hook

Used when personally identifiable data present. Can be more easily rid of biases by design. Disadvantage: sometimes less precise than real data. May differ from real data in non-obvious ways.

Definition

Accurate, unbiased, well-structured, cleaned, suitable for training.

Memory Hook

Improves accuracy and efficiency of AI models. Requires consideration in data strategy. Synthetic data and AI readiness linked. Data quality is foundational.

Definition

How long data is kept and when deleted.

Memory Hook

Complies with regulations. Routine deletion protects individuals. Keeps data accurate. Expiry date is good security practice. Personal data should not be kept forever.

Rights

Access, rectification, erasure, restriction, portability, objection.

Memory Hook

Article 17 = erasure. Article 33 = breach notification. Article 6 = lawful basis. Article 22 = automated decision-making. Extraterritorial scope.

Scope

Applies to organizations outside EEA offering services to EEA data subjects.

Memory Hook

Made GDPR hugely effective globally. Inspired other countries' privacy legislation. Organizations improved standards everywhere. Better standards for all users.

Definition

California Consumer Privacy Act (2018).

Memory Hook

Most comprehensive US state privacy law. Rights: know, access, delete, opt out of sale, non-discrimination. Applies to businesses doing business in California. Inspired other state laws.

Definition

Biometric Information Privacy Act (Illinois, 2008).

Memory Hook

Regulates collection of biometric identifiers. Statutory damages: \$1,000 per violation, \$5,000 if intentional. Notification, access, deletion, breach notice. Civil remedies available.

Definition

Health Insurance Portability and Accountability Act (1996).

Memory Hook

Protects privacy and security of health information. Applies to health care providers, plans. Requires safeguards for electronic health information. Breach notification required.

Definition

Gramm-Leach-Bliley Act.

Memory Hook

Requires financial institutions to explain privacy practices. Implement security safeguards for customer information. Provide opt-out rights. FTC enforces.

Definition

Financial institutions under FTC jurisdiction must keep customer information secure.

Memory Hook

Requires measures to keep customer information secure. Part of GLBA implementation. Applies to non-bank financial institutions.

Definition

Governs federal agencies' collection and use of personal data.

Memory Hook

Prohibits disclosure without written consent. Limited exceptions (Census Bureau). System of records requirements. Applies only to federal agencies.

Definition

Targets large online platforms with gatekeeper status.

Memory Hook

Bans self-preferencing. Requires interoperability. Transparency obligations. No tying and bundling. Open access to data.

Definition

Requires platforms to prevent illegal content, be transparent, address disinformation.

Memory Hook

Protect against algorithmic bias. Allow opt-out of personalized content. Applies to online intermediaries and platforms. 19 companies currently beholden.

Definition

Prohibit illegal content, prevent discriminatory content, protect privacy.

Memory Hook

Published by Cyberspace Administration of China. Effective August 15, 2023. Applies to text, pictures, sounds, videos. Not infringe on others' rights.

Definition

Personal Information Protection Law (China).

Memory Hook

Mandates data minimization, user consent, algorithmic transparency. Comprehensive personal information law. Similar to GDPR in scope.

Definition

CPA (2021).

Memory Hook

Rights: know, access, delete, opt out of sale, opt out of targeted advertising. Data minimization, security, transparency. Applies to businesses in Colorado.

Definition

Consumer Data Protection Act (2021).

Memory Hook

Applies to businesses controlling/processing data of 100,000+ Virginia consumers. Or 25,000+ with >50% revenue from data sales. Rights similar to CCPA.

Definition

Personal Information Protection Act (2005).

Memory Hook

Reasonable security, breach notification within 24 hours. Privacy notice, access, correction, opt-out. Applies to businesses with Illinois resident data.

Date

August 1, 2024.

Memory Hook

Ban on unacceptable risk from February 2, 2025. Most provisions from August 2, 2026. European AI Office enforces. First body to enforce binding AI rules.

Definition

Responsible for compliance, implementation, enforcement of EU AI Act.

Memory Hook

First body in world to enforce binding AI rules. Established by EU AI Act. Sets global standards.

Requirements

Record keeping, impact assessment, transparency, human oversight.

Memory Hook

Providers must keep records of datasets, methodologies, oversight. Impact assessment on fundamental rights. Notify when interacting with AI. Label deepfakes.

Prohibited

Social scoring, biometric categorization, emotion recognition in workplace/education, untargeted facial scraping, manipulation.

Memory Hook

Six months for companies to comply. Fines up to 35M euros or 7% of global turnover. Based on offense and company size.

Range

7.5M euros (1.5%) to 35M euros (7%) of global turnover.

Memory Hook

Depends on offense and company size. Proportional to severity. First body to enforce binding AI rules. Sets global precedent.

Date

October 30, 2023.

Memory Hook

President Biden. Safe, Secure, and Trustworthy Development and Use of AI. Requires safety test results sharing with government. Addresses biases, algorithmic decision-making, cybersecurity threats. AI Safety and Security Board established by DHS (April 26, 2024).